

Chapter 2: Signal Theory

Exercises – Solutions

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Exercise 1:

The continuous-time signal $x(t)$ is generated according to:

$$x(t) = \sum_{k=1}^3 b_k \cos(k\omega_0 t)$$
$$\text{with } \omega_0 = \frac{\pi}{5} \text{ rad/s,} \quad \text{and} \quad b_k = \begin{cases} \frac{2}{k\pi} & \text{for } k = 1, 3 \\ 0 & \text{for } k = 2 \end{cases}$$

- Is $x(t)$ a periodic signal, and if so, what is its fundamental period?
- Is $x(t)$ an even or odd function? Motivate your answer.
- Determine the Fourier series of $x(t)$ in the complex exponential form. For this purpose, use the formula $\cos(\varphi) = \frac{1}{2}(e^{j\varphi} + e^{-j\varphi})$.
- Determine the power of the signal $x(t)$

Solution

a. $x(t) = b_1 \cos(\omega_0 t) + b_3 \cos(3\omega_0 t)$

The signal $x(t)$ is made up of the sum of two cosines (since $b_2 = 0$). Since these two cosine functions are integer multiples of ω_0 , we can directly conclude that $x(t)$ is periodic and that the fundamental frequency is ω_0 . The fundamental period is then $2\pi/\omega_0 = 10$ sec.

b. $x(t)$ is an even function. Because $\cos(t) = \cos(-t)$ for all t , it follows that $x(t) = x(-t)$.

c. The signal $x(t)$ is given by: $x(t) = b_1 \cos(\omega_0 t) + b_3 \cos(3\omega_0 t)$

with $b_1 = 2/\pi$ and $b_3 = 2/(3\pi)$. It is asked to determine the Fourier series of $x(t)$ in its complex exponential form i.e. to rewrite $x(t)$ as a summation of complex exponentials at harmonics of its fundamental frequency ω_0 :

$$x(t) = \sum_{k=-\infty}^{+\infty} c_k e^{jk\omega_0 t}$$

Consequently, in order to determine the Fourier coefficients, we will just rewrite the cosines into complex exponentials using Euler formula:

$$\begin{aligned} x(t) &= b_1 \cos(\omega_0 t) + b_3 \cos(3\omega_0 t) \\ x(t) &= \frac{b_1}{2} (e^{j\omega_0 t} + e^{-j\omega_0 t}) + \frac{b_3}{2} (e^{j3\omega_0 t} + e^{-j3\omega_0 t}) \end{aligned}$$

The following values for the Fourier coefficients c_k :

$$c_1 = c_{-1} = b_1/2 = 1/\pi$$

$$c_3 = c_{-3} = b_3/2 = 1/3\pi$$

$$c_k = 0 \text{ for all other values of } k$$

d. Using Parseval's theorem, The power of $x(t)$ is given by

$$\sum_{k=-3}^3 |c_k|^2 = 2 \left(\frac{1}{\pi^2} + \frac{1}{9\pi^2} \right) = \frac{20}{9\pi^2}$$

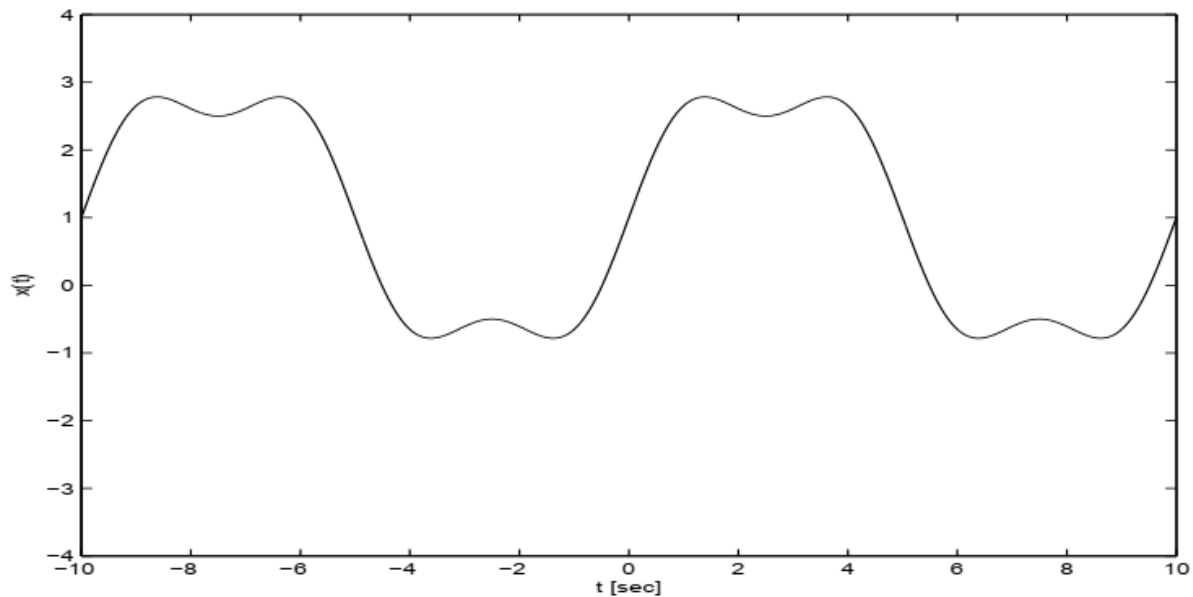
Exercise 2:

1-

In Figure 1, the infinite-time signal $x(t)$ is depicted over the interval $[-10, 10]$. This continuous-time signal $x(t)$ is periodic and is generated as

$$x(t) = a_0 + \sum_{k=1}^3 a_k \sin(k\omega_0 t)$$

with $\omega_0 = 2\pi/10$ rad/s, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$ and $a_3 = 1$



- Is $x(t)$ an odd signal, an even signal or neither of the two? Motivate your answer.
- Determine the fundamental period of $x(t)$ and its fundamental frequency.
- Determine the Fourier series of $x(t)$ in the complex exponential form.
- Determine the power of the signal $x(t)$.

Solution:

- Due to the constant term a_0 , the signal is neither even nor odd.
- Since $x(t)$ is made up of harmonics of ω_0 . The fundamental frequency is thus equal to $\omega_0 = 2\pi/10$ rad/s.

The fundamental period is thus given by: $T = 2\pi/\omega_0 = 10$ seconds. That $x(t)$ has indeed a period of 10 seconds can also be checked in Figure 1.

c. The signal $x(t)$ can be rewritten as:

$$x(t) = a_0 + \sum_{k=1}^3 a_k \sin(k\omega_0 t)$$

$$x(t) = a_0 + \sum_{k=1}^3 \frac{a_k}{2j} (e^{jk\omega_0 t} - e^{-jk\omega_0 t})$$

$$c_{-3} = -a_3/2j = j/4 ,$$

$$c_3 = a_3/2j = -j/4 ,$$

$$c_{-2} = c_2 = 0,$$

$$c_{-1} = -a_1/2j = j,$$

$$c_1 = a_1/2j = -j,$$

$$c_0 = a_0 = 1 \text{ and}$$

$$c_k = 0 \text{ for all other } k.$$

d- Using Parseval's theorem, The power of $x(t)$ is given by

$$\sum_{-3}^3 |c_k|^2 = 2\left(\frac{1}{16} + 1\right) + 1 = \frac{25}{8} = 3.125$$